NII-UIT at VBS2025: Multimodal Video Retrieval with LLM Integration and Dynamic Temporal Search

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Abstract. In summary, our innovative retrieval system for interactive video search, developed for the VBS 2025 competition, significantly elevates the user experience through the utilization of LLMs. By integrating LLMs for advanced query expansion techniques, we effectively address ambiguities and broaden search parameters, resulting in enhanced retrieval accuracy. Our proposed multimodal search framework is designed to support a variety of input types, including text queries, visual data, object filtering, and visual queries generated with Stable Diffusion, providing users with a flexible and intuitive search experience. Furthermore, our dynamic temporal search strategy offers a comprehensive evaluation of frame relevance, surpassing traditional methods and delivering a richer and more effective video retrieval experience for users.

Keywords: Interactive video search · Temporal search

1 Introduction

The Video Browser Showdown (VBS) [15] is an annual international competition, held at the International Conference on Multimedia Modeling (MMM), that showcases the latest innovations in interactive video retrieval technologies. For over 13 years, VBS has featured a variety of tasks, including Visual and Textual Known-Item Search (KIS-V, KIS-T), Ad-Hoc Video Search (AVS), and, more recently, Question Answering (Q/A). In KIS tasks, participants locate a specific frame or segment, while in AVS, they retrieve as many relevant segments as possible matching a description. Introduced in 2024, the Q/A task requires answering questions based on video content. For VBS 2025, participants will search through large datasets such as V3C [13], LapGynLHE, and Marine Video Kit (MVK) [14]. The competition is divided into expert and novice sessions,

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with the latter inviting audience volunteers, highlighting the need for systems that are both high-performing and user-friendly.

This paper presents an approach that leverages Large Language Models (LLMs) to enhance query expansion in video search tasks. By using LLMs, our system interprets and enriches user queries through paraphrases, synonyms, and contextually relevant variations, improving the retrieval of relevant video segments and addressing ambiguities. This innovation enhances both precision and relevance in search results.

We employed a robust multi-modal retrieval system that integrates advanced Vision-Language Models (VLMs) in each key module, ensuring high performance in both text and visual search while handling cross-modal queries with accuracy and flexibility. Additionally, we incorporate query expansion, image generation using Stable Diffusion, as well as object counts and spatial positioning to refine search results. All of these advanced features are packaged in a user-friendly interface, enabling both novice and expert users to conduct sophisticated searches with ease.

Traditional methods tend to oversimplify the complexity of temporal relationships by classifying events strictly as either preceding or following one another [5]. However, in interactive video search and retrieval, accurately interpreting the temporal sequence of consecutive descriptions is essential, particularly when the order of events is unclear. This simplification can reduce the effectiveness of these methods in scenarios where a more nuanced understanding is needed. To address this issue, we propose a dynamic temporal search approach that assesses frame relevance both before and after the previous result. This approach is designed to improve search performance, especially when the temporal relationship between descriptions is ambiguous.

2 Proposed System

The NII-UIT retrieval system, depicted in Fig. 1 utilizes a two-phase pipeline consisting of offline and online stages.

In the offline phase, keyframes are extracted from each shot. Feature vectors of each keyframe are generated and stored in a vector database Milvus [1]. These keyframes are also processed using an object detection model, with the detected objects being stored in a tabular database.

During the online phase, users can submit either visual or textual queries. Textual queries can be paraphrased using models like GPT-4o or used as prompts for generative models to produce images. In the case of a Q/A task, the user's question can be transformed into a textual description. The results from different query types are then aggregated and filtered based on object query constraints to provide the final output.

[1] https://milvus.io/.

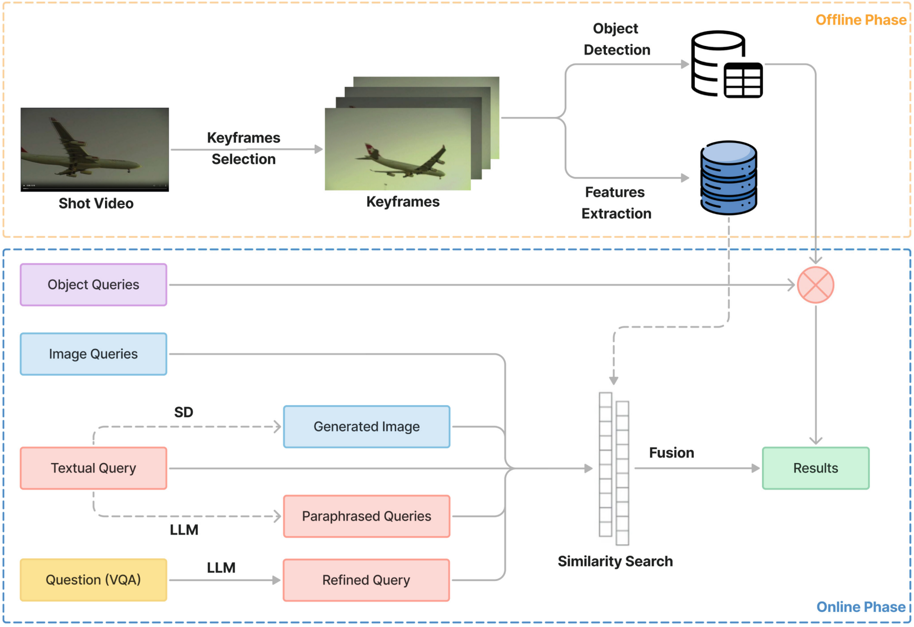

Fig. 1. The proposed system operates in two stages: offline and online. In the offline stage, datasets are processed and indexed for efficient querying. In the online phase, the system analyzes user queries, searches the database, and presents relevant results.

2.1 Pre-processing

VBS presents unique challenges, particularly due to the large datasets involved in retrieval tasks. Processing every frame of video sequences is highly inefficient, leading to increased processing time, storage demands, and slower retrieval speeds. To address this, we developed an optimized keyframe selection strategy inspired by Vibro [6], enhanced with BEiT-3 [16] for semantic feature extraction.

Our method uses BEiT-3 to extract semantic features from every tenth frame, retaining only those with significant differences. Keyframes are saved in WebP format to reduce file size while preserving quality, improving processing speed without affecting retrieval performance.

2.2 Embedding-Based Retrieval

Recent advances in cross-modal VLMs, which embed images and text in a shared latent space, have driven significant progress in image and text retrieval tasks. Notably, VISIONE 5.0 [1], the winner of the last VBS competition, integrates models such as OpenCLIP ViT-L/14 [7], CLIP2Video [4], and ALADIN [10], demonstrating the power of VLM-based approaches. Other models, including BEiT-3 [16] and OpenCLIP H-14 [3], have further advanced the field. BEiT-3 enhances image-text alignment by leveraging Multiway Transformers, masked "language" modeling across modalities, and an effective scaling strategy, while OpenCLIP H-14 leverages a large pre-training dataset for better retrieval accuracy. The InternVL-G model [2], combining InternViT and QLLaMA, set new performance records on COCO [8] and Flickr30K [11].

In our system, we leverage the capabilities of advanced VLMs to enhance both textual and visual search functionalities. The retrieval results from each model are first normalized and then combined using a data fusion algorithm, allowing us to effectively balance their strengths and weaknesses and deliver a more robust and comprehensive search experience.

2.3 Query Expansion

In video search tasks, users frequently provide brief, vague, or incomplete queries that do not fully represent their information needs. This lack of detail often results in less accurate search results, reducing retrieval effectiveness.

To tackle this issue, we employ query expansion techniques to enhance the depth and relevance of user queries. Our approach leverages the capabilities of an LLM such as GPT-4o to automatically generate multiple paraphrased versions of the original query, expanding its interpretive range and potential meanings. This method offers users five alternative rephrasings, each providing a distinct variation that may more accurately reflect their intent. By allowing users to select the version that best aligns with their search objectives, we can improve the search process's effectiveness. Additionally, these paraphrased queries can be executed in parallel, enabling users to view either individual search results for each version or a final ranked list that fuses the outcomes, providing a comprehensive and tailored search experience.

2.4 Visual Query Generation

Recent advancements in generative models, particularly Stable Diffusion (SD) [12], have demonstrated remarkable abilities to generate highly detailed images from textual descriptions. Zhixin Ma et al. [9] underscore the potential of utilizing visual queries created by generative models like SD to improve performance in retrieval tasks.

Building on this approach, our system leverages SD to generate images that serve as visual queries, enhancing the retrieval process by providing rich, image-based representations of search intent. Similar to query expansion techniques, our system enables users to explore results based on a single generated image or combine retrieval outcomes from multiple images. This approach introduces greater flexibility in search, allowing for more comprehensive and accurate retrieval results by enriching the query with multiple visual perspectives.

2.5 Object Filtering

While embedding-based search models excel at retrieving relevant content, they struggle with object counting and distinction in complex video environments due to challenges like object overlap, occlusion, and scale variations. To overcome these, we integrated Co-DETR [17], a model optimized for precise object detection and counting. Co-DETR accurately identifies all COCO objects in each video keyframe, addressing the limitations of embedding models. By utilizing Co-DETR's data, we filter keyframes based on query-specific objects, enhancing retrieval relevance and accuracy in complex visuals.

2.6 Multi-modal Search and Dynamic Temporal Search

For multi-model search, users can seamlessly integrate various search modalities such as textual and visual queries, object-based filtering, query expansion, and image generating resulting in a single-stage result. The scores for each shot in the ranked lists derived from textual, visual, query expansion, and image generation components are first normalized and then aggregated using mean pooling. Once aggregated, the object filtering module is applied to exclude shots that do not contain the specified objects. The remaining shots are reranked and returned to the UI. To versatilize the searching experiment, we also integrate temporal search within our system, which allows for an unlimited number of single stages, enabling the system to refine and adapt queries over time iteratively.

For KIS-T queries, where the temporal relationship between descriptions is unclear and it is ambiguous whether additional information pertains to events before or after the initial description. Therefore we employ a specialized strategy to integrate the results of consecutive descriptions. Specifically, inspired by Vitrivr [5], we have developed an enhanced temporal search mechanism that explores the shots surrounding the initial result, rather than simply determining if the next stage occurs before or after, as per traditional methods. These surrounding shots are assessed for their relevance to the new query information, and we rerank the results by aggregating scores across these stages.

2.7 Question Answering - Q/A

The Q/A task introduced at VBS 2024 presents challenges, requiring teams to provide precise answers to specific questions about video content. Interestingly, these questions can also serve as useful cues for video retrieval. For example, one of the Q/A tasks at VBS 2024 asks, "*There is also a red Volkswagen Golf GTI at this show. What is the license plate number*?" This can be transformed into a textual description, such as "*A red Volkswagen Golf GTI with a license plate*," either manually or with the help of an LLM. This refined version can then be integrated using temporal search 2.6 approach for KIS-T.

2.8 User Interface

Our web-based application provides an intuitive interface, displaying retrieval results in two modes: one groups near shots for easy comparison, and the other highlights the top-scoring frame from each shot, helping users quickly find the most relevant content.

Figure 2 illustrates the interface with the Advanced Mode activated. This mode provides users with additional options, such as adjusting model weights, paraphrasing queries, or generating visual queries from textual descriptions. For novice users, Advanced Mode is disabled to help them focus more on search without the complexity of additional settings.

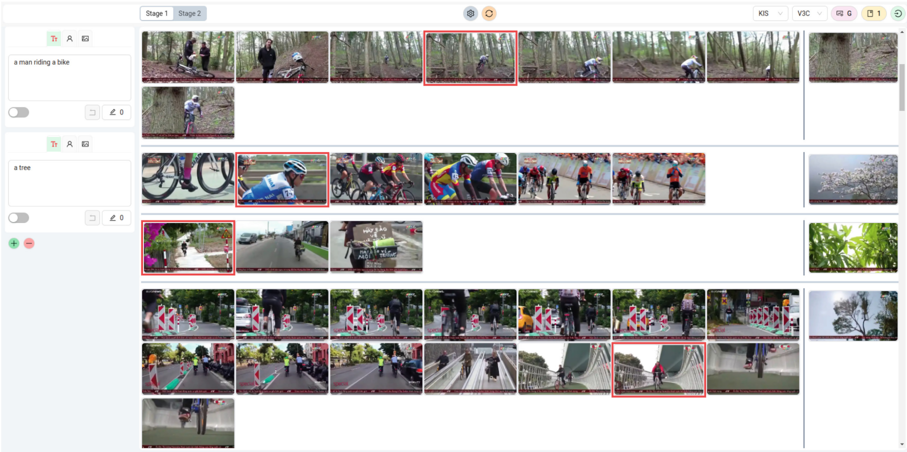

Fig. 2. The UI features two main panels: the left offers search options like visual/text queries, paraphrasing, visual query generation, object filtering, and temporal adjustments. The right shows results, grouping near shots for comparison, with temporal matches on the far right. A top bar lets users switch modes, adjust models, weights, and refine queries.

3 Conclusion

In conclusion, our innovative video search system for VBS 2025 enhances user experience by utilizing LLMs for query expansion, addressing ambiguities and improving retrieval accuracy. Our multimodal framework supports diverse input types, including text and visual queries, object filtering, and visual queries generated with SD, offering flexibility and ease of use. Additionally, our dynamic temporal search strategy surpasses traditional methods, providing a more comprehensive and effective evaluation of frame relevance for an enriched video retrieval experience.

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